

Reviewer Pack — Minimal Tropical Symplectic Index and Communication Complexi...

Entry 5f43f7f0aef3 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-06-02 00:35:02 UTC

Minimal Tropical Symplectic Index and Communication Complexity Rank Correlation

Entry ID: 5f43f7f0aef3 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-02 00:35:02 UTC

1. Conjecture

Field A (mathematical branch): Tropical Geometry × Symplectic Geometry **Field B** (complexity object): Communication Complexity

Statement:

For any given d -regular graph G , the minimal tropical symplectic index ($\text{tsi}(G)$) of its associated Tseitin formula φ_G is linearly correlated with its communication complexity rank $r(\varphi_G)$, such that $\text{tsi}(G) = \Theta(r(\varphi_G))$. Moreover, for all instances with $n \leq 40$, $|\text{tsi}(G) - r(\varphi_G)| \leq 0.1$.

Rationale (proposer's reasoning):

Tropical symplectic geometry offers a rich algebraic structure that can be applied to the analysis of complexity classes, potentially revealing underlying geometric properties related to communication complexity. By establishing a link between this new invariant and communication complexity rank, we could gain insights into the fundamental nature of distributed computation.

Taxonomy category: TROPICAL_SYMPLECTIC_INDEX (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 01a995559c8ef4c0

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if the mean absolute difference between $\text{tsi}(G)$ and $r(\varphi_G)$ across at least 80% of the seeds (24 out of 30) is less than or equal to 0.1, and the Pearson correlation coefficient across all seeds is greater than 0.7.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	1.00	SAFE	SAFE
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	1.00	SAFE	SAFE
KARP_LIPTON	SAFE	1.00	SAFE	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=0, elapsed=0.0s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def gaussian_elimination(A):
        m, n = len(A), len(A[0])
        for i in range(m):
            max_row = i + max(range(i, m), key=lambda j: abs(A[j][i]))
            A[i], A[max_row] = A[max_row], A[i]
            if A[i][i] == 0:
                continue
            for j in range(n):
                A[i][j] /= A[i][i]
            for k in range(m):
                if k != i and A[k][i] != 0:
                    factor = A[k][i]
                    for j in range(n):
                        A[k][j] -= factor * A[i][j]
        return A

    def matrix_multiply(A, B):
        m, n, p = len(A), len(B), len(B[0])
        C = [[0] * p for _ in range(m)]
        for i in range(m):
            for j in range(p):
                for k in range(n):
                    C[i][j] += A[i][k] * B[k][j]
        return C

    def determinant(A):
        m, n = len(A), len(A[0])
        if m != n:
            raise ValueError("Matrix must be square")
        if n == 1:
            return A[0][0]
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```

det = 0
for j in range(n):
    submatrix = [row[:j] + row[j+1:] for row in A[1:]]
    det += (-1) ** j * A[0][j] * determinant(submatrix)
return det

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        if A[i][i] == 0:
            continue
        for j in range(n):
            A[i][j] /= A[i][i]
        for k in range(m):
            if k != i and A[k][i] != 0:
                A[k] -= A[k][i] * A[i]
# ... [truncated, full source in replay tarball]

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

me': 'tsi(G)', 'metric_value': None, 'instances_tested': 1, 'n_max': 27, 'conjecture_holds': False, '
TRIAL: {'metric_name': 'tsi(G)', 'metric_value': None, 'instances_tested': 1, 'n_max': 21, 'conjectur
TRIAL: {'metric_name': 'tsi(G) - r(φ_G)', 'metric_value': 0, 'instances_tested': 1, 'n_max': 18, 'con
TRIAL: {'metric_name': 'tsi(G) - r(φ_G)', 'metric_value': 0, 'instances_tested': 1, 'n_max': 15, 'con
TRIAL: {'metric_name': 'tsi(G) - r(φ_G)', 'metric_value': 0, 'instances_tested': 1, 'n_max': 7, 'conj
TRIAL: {'metric_name': 'tsi(G)', 'metric_value': None, 'instances_tested': 1, 'n_max': 25, 'conjectur
TRIAL: {'metric_name': 'tsi(G)', 'metric_value': None, 'instances_tested': 1, 'n_max': 23, 'conjectur
TRIAL: {'metric_name': 'tsi(G) - r(φ_G)', 'metric_value': 0, 'instances_tested': 1, 'n_max': 9, 'conj
TRIAL: {'metric_name': 'tsi(G) - r(φ_G)', 'metric_value': 0, 'instances_tested': 1, 'n_max': 38, 'con
TRIAL: {'metric_name': 'tsi(G) - r(φ_G)', 'metric_value': 0, 'instances_tested': 1, 'n_max': 16, 'con
TRIAL: {'metric_name': 'tsi(G) - r(φ_G)', 'metric_value': 0, 'instances_tested': 1, 'n_max': 36, 'con
TRIAL: {'metric_name': 'tsi(G)', 'metric_value': None, 'instances_tested': 1, 'n_max': 19, 'conjectur
TRIAL: {'metric_name': 'tsi(G) - r(φ_G)', 'metric_value': 0, 'instances_tested': 1, 'n_max': 17, 'con
RESULT: FALSIFIED counterexample="mapping_undefined" first_failing_seed=11

```


8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

The test code contains multiple definitions of the same functions, which suggests a bug in the code that may not be calculating the intended mathematical operations correctly. Additionally, the test only considers up to $n=15$, which is too small to confirm a linear correlation over a larger range of graph sizes.

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge_falsified | original: The test results indicate that the conjecture does not hold for at least one seed (first_failing_seed=11) due to a 'mapping_undefined' error, which is | next: Investigate the cause of the 'mapping_undefined' error in the test code. Extend the test to include a larger range of graph sizes ($n > 15$) and retest the conjecture. Additionally, address any potential bugs in the test code that may affect the accuracy of the calculations.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	48800
2	preregistration	ollama_remote	glm4:latest	0	0	9712
3	novelty	ollama_remote	glm4:latest	0	0	8499
4	novelty	ollama_remote	glm4:latest	0	0	20962
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	19128
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	37213
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	16943
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	105908
9	critic	ollama_remote	glm4:latest	0	0	18084
10	judge	ollama_remote	glm4:latest	0	0	15241

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 300490 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in `research/audit/5f43f7f0aef3.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/5f43f7f0aef3.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/5f43f7f0aef3.tar.gz` (if generated)